Network Influence Analysis Framework

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1 Abstract

To create a framework for the analysis of influence within networks, particularly the investigation of heuristics for finding the most influential nodes in an efficient manner.

2 Problem Discussion

Network Influence Analysis relates to studying social networks to determine the set of k nodes which have the most 'influence', or greatest spread of information through the network, in the network. Using brute force is incredibly time consuming on the order of $O\left(\binom{|V|}{k}\right)$. Using certain heuristics such as the degree of a node to choose the seed nodes is significantly more efficient running in polynomial time, I will focus on allowing for simple implementations and analysis of different heuristics.

There is a significant amount of information out there regarding how these models are implemented and computed, there are graph packages in most programmming languages which make it relatively simple to implement them as described.

A typical piece of research in this area would be to attempt to find a new heuristic to find the set of seed nodes which have a greater influence than other heuristics. To do this the researcher would need to implement the model, implement representing their chosen networks in their model, implement the known heuristics and their new hypothesised heuristic and analyse the influence spread of each of them to be able to see if there is a measurable difference.

Creation of a framework for performing these tasks would allow for creating such a study to be simpler, quicker and more reproducible due to a consistent environment.

3 Objectives

- Import and Export of different graph formats
- Generation of synthetic graphs (**Kleinberg, Eppstein**)
- Independent Cascade Model (**IC**)
- Linear Threshold Model (**LTM**)
- Decreasing Cascade Model (**DC**)
- Different Sampling of large graphs to make them manageable
- Different heuristics to find seed nodes
- Simple Creation of new sampling methods
- Simple Creation of new heuristics
- Python **whl** and **tarball** packaging to allow easy installation

4 Methodology

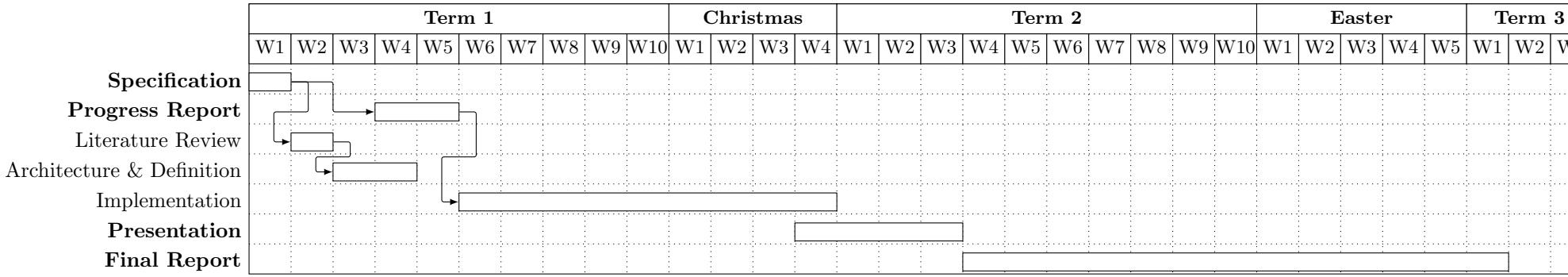
The project will be broken up into three main stages; research, implementation and evaluation.

The research stage will require a detailed literature review of articles pertaining to network influence modelling, methods, heuristics and sampling. I will identify the validity and applications of each source and find if there are any gaps in the presented knowledge to research further.

The implementation stage of the project will require a methodology that can be adapted to changing requirements or a shift in focus based on information learnt later in the research phase. I will use an agile-like methodology, of one to two week segments determined by meetings with my supervisor with any changes to requirements determined during these meetings.

The final stage will be to use my framework to replicate the results of the researched papers, showcase how to create new sampling methods and heuristics and how to compare the results of different sampling, heuristics and models on different networks.

5 Timelines


Term 1
Christmas
Term 2
Easter
Term 3
W1 W2 W3 W4 W5 W6 W7 W8 W9 W10 W1 W2 W3 W4 W1 W2 W3 W4 W5 W6 W7 W8 W9 W10 W1 W2 W3 W4 W5 W1 W2 W
Specification
Progress Report
Literature Review
Architecture & Definition
Implementation
Presentation
Final Report

6 Resources and Risks

7 Legal, Social, Ethical and Professional Considerations